

Assessment of the $^{49}\text{Ti}(\text{d},\text{x})$ route to produce the theranostic radionuclide ^{47}Sc based on cross-section measurements

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Abstract. The availability of the theranostic radionuclide ^{47}Sc is hindered by the lack of a reliable high purity production method. The aim of this work is to evaluate a less considered pathway exploiting deuteron beams on enriched ^{49}Ti targets. The investigation started from the measurements of the production cross sections of the emergent radionuclide ^{47}Sc , as well as of the co-produced contaminants which play an important role on the final purity and in the evaluation of the dose eventually administered to a patient. Enriched ^{49}Ti targets, manufactured at the INFN-LNL (National Institute of Nuclear Physics-Legnaro National Laboratories), were activated in several irradiation runs carried out at the GIP ARRONAX (Group of Public Interest Accelerator for Research in Radiochemistry and Oncology in Nantes-Atlantique) facility cyclotron utilizing the tunable 15÷35 MeV deuteron beam. The obtained cross section trends, which constitute the first measurement for some radionuclides, are compared to theoretical curves and literature data, when available. The experimental cross sections are then discussed to assess $^{49}\text{Ti}(\text{d},\text{x})$ as a possible production route for ^{47}Sc . The results demonstrated that $^{49}\text{Ti}(\text{d},\alpha)^{47}\text{Sc}$ can be considered as a valid alternative production route for the supply of this medically interesting radionuclide.

Keywords. Production cross sections, deuteron-induced nuclear reactions, theranostic scandium, ^{47}Sc , enriched ^{49}Ti targets, cyclotron-based production

1. Introduction

As a potential theranostic radionuclide, ^{47}Sc has drawn attention in recent years thanks to its emission of low-energy β^- with a mean energy of 162.0 keV (NNDC website, 2025). This energy is very similar to the β^- emission of the gold-standard ^{177}Lu (mean energy 133 keV), making it a valid alternative for the internal radiation therapy of small and metastatic tumours [De Nardo, 2025]. However, the key advantage of ^{47}Sc is its possibility to be paired to a PET (Positron Emission Tomography) diagnostic counterpart, namely ^{43}Sc or $^{44\text{g}}\text{Sc}$, both β^+ emitters. This configuration is identified as true theranostic pair, enabling a more precise assessment of the treatment plans based on the patient's response at the disease imaging stage. Nowadays, the most common pair is $^{68}\text{Ga}/^{177}\text{Lu}$, a couple with similar but not identical

chemical properties. In fact, they have different coordination geometries and chemical structures, resulting in two radiopharmaceuticals with different stability and biodistribution. Thus, the use of true theranostic pairs, such as $^{43}\text{Sc}/^{47}\text{Sc}$ or $^{44\text{g}}\text{Sc}/^{47}\text{Sc}$, will constitute an improvement of clinical procedures. Moreover, ^{43}Sc and $^{44\text{g}}\text{Sc}$ have the potential advantage to have a longer half-life than ^{68}Ga (3.891 h/4.0420 h vs. 1.1285 h) which can better match biomolecules with a slower biodistribution profile. Additionally, ^{47}Sc has a γ emission of ideal energy and suitable intensity (see table 1) to be used with SPECT (Single Photon Emission Computed Tomography) cameras to perform pre- and post-therapeutic imaging to monitor the treatment effectiveness.

First step for the establishment of the ^{47}Sc production chain consists in individuating the most convenient nuclear reaction (one or more) for a medically suitable final product [Pupillo, 2022]. The main aspects to be considered are the quantity of the radionuclide of interest produced, ^{47}Sc , as well as the presence of co-produced contaminants, especially the isotopic ones with comparable or longer half-lives (like ^{48}Sc and ^{46}Sc), since cannot be chemically separated. Those co-products can degrade the quality of the final product, evaluated in term of radionuclidic purity for which a conventionally acceptable threshold value of 99% is considered in the context of this work. In last decades, many different production pathways have been investigated, involving cyclotron facilities, nuclear reactors, and photoproduction with electron linear accelerators [Mikolajczak, 2021].

Both fast (nuclear reaction: $^{47}\text{Ti}(\text{n,p})^{47}\text{Sc}$) and thermal (generator system: $^{46}\text{Ca}(\text{n},\gamma)^{47}\text{Ca} \rightarrow ^{47}\text{Sc}$) neutrons were investigated for the production at nuclear reactors. The latter demonstrated significant advantages in the elevated radionuclidic purity and higher yield, counterbalanced by the cost, influenced by enrichment and availability, of the ^{46}Ca enriched material with respect to the ^{47}Ti [Domnanich, 2017]. However, the development of a recovery method can minimize this drawback. Nevertheless, the issue about the availability of nuclear reactors, especially the ones delivering fast neutrons, remains.

The use of bremsstrahlung radiation for the photoproduction of ^{47}Sc was studied with three different natural target materials: titanium (reaction of interest: $^{48}\text{Ti}(\gamma,\text{n})^{47}\text{Sc}$), vanadium ($^{51}\text{V}(\gamma,\alpha)^{47}\text{Sc}$), and calcium ($^{48}\text{Ca}(\gamma,\text{n})^{47}\text{Ca} \rightarrow ^{47}\text{Sc}$). The limiting factor of these production methods is the low amount of produced activity while maintaining an adequate purity. Several options can be explored to overcome this downside. The use of enriched target materials is a possibility which can increase the yield, as well as the purity since the co-production of Sc-contaminants is limited [Loveless, 2019], particularly in the case of calcium targets. In fact, $^{\text{nat}}\text{V}$ is composed by ^{51}V at 99.750%, and $^{\text{nat}}\text{Ti}$ is composed by ^{48}Ti at 73.72%, and, consequently, approaching the 100% enrichment level will not improve significantly the amount of produced activity. Contrariwise, given the low natural abundance of ^{48}Ca (0.187%), the required enrichment level and the limited availability translate in elevated costs [Mikolajczak, 2021]. Alternatively, the increase of the involved masses of target materials can support a more fruitful production, but its mass-dependent behaviour is not deeply studied yet [Snow, 2021]. Moreover, the cost-effective aspect of combining material enrichment with larger mass should be considered. In the end, higher end-point photon energies rise the produced yield but at the expense of the purity due to reach of the thresholds of the contaminants' nuclear reactions.

Charged-particle-induced production can rely on proton, deuteron, and alpha beams. Alpha particles have been exploited through the $^{44}\text{Ca}(\alpha, p)^{47}\text{Sc}$ reaction. It was assessed that low energies (<20 MeV) are necessary to keep the purity high, resulting in a low physical yield [Aikawa, 2022]. On the other side, two target materials demonstrated to be suitable for the ^{47}Sc production with proton beams. Recently, production studies involving $^{\text{nat}}\text{V}$ [Cingoranelli, 2024] and enriched $^{50}\text{TiO}_2$ targets [Dellepiane, 2022; Cingoranelli, 2023] showed the validity of this production route reporting highly pure products (radionuclidic purities $> 99.5\%$) with activity quantities satisfying the needs for at least preclinical trials. In this work, instead, the alternative production using deuteron beams is considered. The advantage of using deuteron beams can be the need of less target material due to the deuterons shorter range in comparison to protons at a given energy, which is a positive aspect especially when enriched expensive materials are involved. Moreover, a pure ^{47}Sc production is expected thanks to the (d, α) channel which typically opens faster than others, at low energies. Cross section measurements of the $^{49}\text{Ti}(d, x)^{47,46,48}\text{Sc}, ^{48}\text{V}$ nuclear reactions, performed in the context of the SPES_MED project funded by CSN3 of INFN (National Institute of Nuclear Physics), are presented. Only $^{49}\text{Ti}(d, x)^{47,46}\text{Sc}$ trends were already measured in the past [Chen and Miller, 1964], and so, thanks to this study, the required knowledge to evaluate this production pathway is extended.

1. Material and Methods

In the experiments dedicated to the investigation of the ^{47}Sc production with deuteron beams, enriched ^{49}Sc targets, previously used in proton-induced nuclear reactions measurements [De Dominicis, 2025], were recycled. Approximately, three years of cooling elapsed to assure that all the radionuclides expected to be produced in the new irradiations (^{48}V and Sc-isotopes reported in table 1) were almost completely decayed from the previous ones. However, preliminary γ spectroscopy measurements were performed to evaluate the presence of eventual residual activities of longer-lived radionuclides previously produced. Some details about target manufacturing are here reported in brief since they were already described in previous works [Mou, 2024; Cisternino, 2023; Cisternino, 2022; Skliarova, 2020]. The targets were manufactured at the INFN-LNL (Legnaro National Laboratories) starting from metallic ^{49}Ti powder purchased from the Oak Ridge National Laboratories (Tennessee, USA) with the enrichment level reported in table 2. Due to its spongy form and variable dimensions of the grains (from few μm up to 2 mm), the powder needed a cryomilling process [Cisternino, 2023] before exploiting the HIVIPP (High Intensity Vibrational Powder Plating) technique to obtain a thin Ti deposition on a $^{\text{nat}}\text{Al}$ substrate [Cisternino, 2022; Skliarova, 2020]. EBS (Electron Backscattering Spectroscopy) analysis, performed at the AN2000 accelerator sited at INFN-LNL, were then performed to determine the thickness of the deposited Ti layers [Mou, 2024].

Radionuclide	Half-life	E of used γ [keV]	I_γ [%]
^{48}V	15.974 d 3	983.525 4 1312.105 6	99.98 4 98.2 3
^{48}Sc	43.71 h 9	1037.522 12	97.5 20
^{47}Sc	3.3492 d 6	159.381 15	68.3 4
^{46}Sc	83.79 d 4	889.277 3	99.984 10
^{61}Cu	3.336 h 10	282.956 10	12.7 20
^{56}Co	77.236 d 26	1238.288 15	66.46 12
^{58}Co	70.86 d 6	810.7593 20	99.450

Table 1. Half-lives, energy and intensity of the γ lines of the produced radionuclides used in cross section calculations [NNDC website, 2025]. The italic numbers are the last digits of the associated uncertainties.

	^{46}Ti	^{47}Ti	^{48}Ti	^{49}Ti	^{50}Ti
^{49}Ti	0.22	0.22	2.71	96.25	0.60
$^{\text{nat}}\text{Ti}$	8.25	7.44	73.72	5.41	5.18

Table 2. Enrichment level of the purchased ^{49}Ti metallic powder. Natural Ti abundances are also reported for comparison.

Four stacks including 3 or 4 ^{49}Ti thin targets were arranged to obtain a maximum of 15 cross section experimental values for each radionuclide. Each stack was bombarded at a different deuteron impinging energy ranging from 16 to 30.7 MeV. In each stack, besides the material of interest ^{49}Ti , $^{\text{nat}}\text{Ni}$ foils (25 μm) were inserted to monitor the beam intensity through the IAEA (International Atomic Energy Agency) recommended reactions $^{\text{nat}}\text{Ni}(p,x)^{61}\text{Cu}$, $^{56,58}\text{Co}$ (Monitor reactions website, 2024; Hermanne, 2018). Following the $^{\text{nat}}\text{Ni}$ in the stack, thin $^{\text{nat}}\text{Al}$ foils (10 μm) were included as catchers of the nuclei escaping from $^{\text{nat}}\text{Ni}$ foils due to their residual recoil kinetic energy. In addition, thicker $^{\text{nat}}\text{Al}$ layers (100÷750 μm) have been used as beam energy degraders between one ^{49}Ti sample and the following one. All the used materials had 99% as minimum purity, and were purchased from the Goodfellow Cambridge Ltd. (Huntingdon, UK). All the listed layers were cut into discs of 12 mm diameters to fit the target-holder, which was then placed in front of the beam. Irradiation runs took place at the GIP ARRONAX facility (Nantes, France) [Haddad, 2008] with a duration between 40 min and 1 h, and a mean beam intensity in the interval 80÷100 nA.

Following the bombardments, a proper cooling time of few hours was waited before performing γ spectroscopy measurements through four different HPGe (Hyper Pure Germanium) detectors available at GIP ARRONAX, which were previously calibrated with point-like certified ^{241}Am and ^{152}Eu sources from CERCA-LEA (Tricastin, France). The activity in each ^{49}Ti , $^{\text{nat}}\text{Ni}$ and thin $^{\text{nat}}\text{Al}$ foil was measured repeatedly in the days after the EOB (End Of Bombardment) to follow the produced radionuclides decay and/or to check for the eventual presence of interferences in the γ peaks. ^{49}Ti and $^{\text{nat}}\text{Ni}$ foils underwent also a long acquisition about 1 month after the EOB to measure the long-lived radionuclides, namely ^{46}Sc , ^{48}V , ^{56}Co , and ^{58}Co , when all the other short-lived co-products have decayed. Each γ spectrum acquisition lasted between 30 min (e.g. the first ones soon after the EOB to measure short-lived radionuclides), and 14 h, (e.g., the overnight and distant from EOB ones which allowed to have a better statistic of long-lived radionuclides).

Data analysis was performed through the jRadView software developed at INFN-LNL for internal use with which the γ lines of the produced radionuclides (table 1) were identified and the corresponding activity evaluated. The interference in the common peaks, namely 983.525 keV and 1312.105 keV, shared between ^{48}V and ^{48}Sc was considered and the contribution subtracted. The extracted data were used to calculate cross sections and related uncertainties according to the formula proposed by Otuka et al. [Otuka, 2017].

The deuteron beam energy associated to each foil was evaluated throughout simulations of the energy loss starting from the nominal energy with the SRIM2013 Monte Carlo code [Ziegler, 2010]. Thanks to the choice of three different monitor reactions, it was possible to also validate the nominal beam energy with the technique proposed in [Mastren, 2019], but adapted to

deuteron beams. The energy uncertainty was extracted from simulations with SRIM2013 code assuming an initial intrinsic uncertainty of the deuteron beam energy of 0.5 MeV and deriving the energy straggling effect across the stacked-target.

2. Results and Discussion

The evaluation of each utilized beam energy independently from its corresponding beam intensity, exploiting the three different IAEA monitor reactions taking place in the ^{nat}Ni foils, gave positive results indicating that the nominal energy was effectively the energy of the deuterons impinging on the stacked-targets. This endorses the use of the nominal beam energies involved to derive the energies in the enriched ^{49}Ti layers to be associated to the calculated cross sections.

The values obtained from data analysis for the produced radionuclides, namely $^{47,46,48}\text{Sc}$, and ^{48}V , corresponding to the real enrichment level of the target material ^{49}Ti used which is indeed quite close to 100% (table 2), are reported in table 3. They are graphed in figures 1 to 4 together with the theoretical data estimated by TENDL-2025 library [Koning, 2019]. For the ^{47}Sc and ^{46}Sc produced radionuclides the excitation function previously measured by Chen et al. [Chen, 1964] are also indicated for comparison.

E_{beam} (MeV)	$\sigma_{47\text{Sc}}$ (mb)	$\sigma_{46\text{Sc}}$ (mb)	$\sigma_{48\text{Sc}}$ (mb)	$\sigma_{48\text{V}}$ (mb)
4.4±1.6	3.7±1.0	-	-	-
6.9±1.1	18.4±2.1	-	-	-
9.2±1.2	23.7±2.8	-	-	-
10.4±0.8	31.3±3.8	1.3±0.9	0.18±0.05	-
13.2±0.9	22.2±2.4	17.8±2.7	0.31±0.12	-
15.0±0.6	20.8±2.3	34.2±4.1	1.8±0.3	-
17.9±0.7	11.5±1.4	49.6±7.6	1.1±0.3	8.6±1.8
19.0±1.1	11.3±1.1	53.1±5.9	1.6±0.2	23.2±7.3
20.8±0.6	9.7±1.1	63.0±13.0	3.2±0.7	45.7±5.4
22.4±0.9	7.3±0.8	67.8±7.8	1.9±0.2	84.3±10.9
24.5±0.8	7.8±0.9	69.1±7.9	3.8±0.4	136.7±16.4
26.4±0.8	6.7±0.7	67.3±8.3	4.6±0.5	158.6±20.3
28.3±0.6	5.6±0.5	35.6±3.8	6.4±0.5	158.8±14.9
30.1±0.6	7.5±0.8	42.7±5.1	10.1±1.1	186.2±21.5
32.4±0.6	9.0±1.0	40.3±5.4	14.3±1.6	194.6±23.4

Table 3. Cross section results for the $^{49}\text{Ti}(\text{d},\text{x})^{46}\text{Sc}$, ^{47}Sc , ^{48}Sc , and ^{48}V nuclear reactions in the beam energy interval 4.4÷32.4 MeV.

The $^{49}\text{Ti}(\text{d},\text{x})^{47}\text{Sc}$ excitation function is reported in figure 1. It can be observed that TENDL-2025 are in global good agreement with experimental values firstly proposed in this work. Literature data by Chen et al. is overestimating the cross-section peak at low energy by approximately 30%. The reason of this inconsistency is not easy to understand because Chen et al. performed their experiments to demonstrate some nuclear reactions properties and so not detailed insights about experimental methods are discussed. Moreover, there are not given uncertainties associated to their measured values that can be used to evaluate the goodness of their results. Same considerations can be done for the case of the $^{49}\text{Ti}(\text{d},\text{x})^{46}\text{Sc}$ reaction, whose data are represented in figure 2. However, in this case the Chen et al. trend underestimates the results obtained in this work by approximately 50%. As for ^{47}Sc , the theoretical data from

TENDL-2025 reproduce the general trend, but the maximum is slightly shifted towards lower energies.

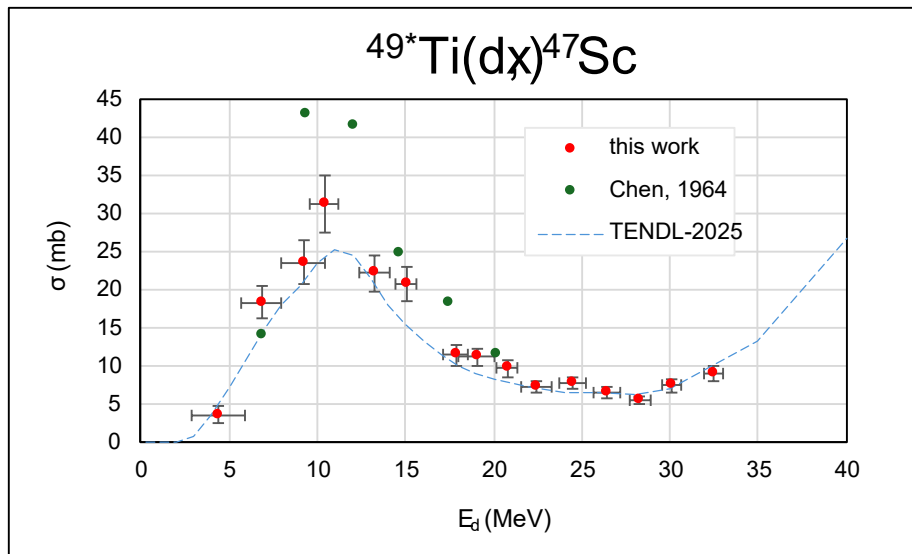


Figure 1. Cross section trend of the $^{49}\text{Ti}(d,x)^{47}\text{Sc}$ reaction (*for target real enrichment).

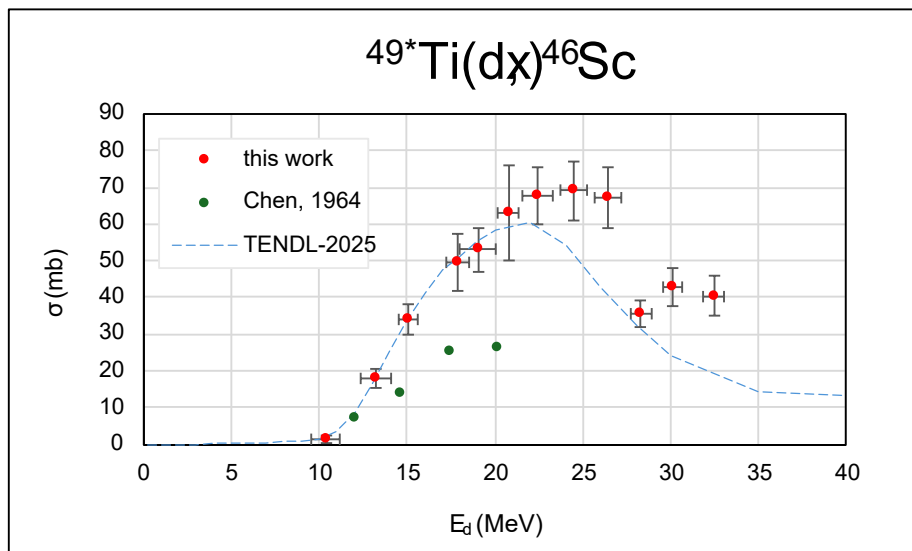


Figure 2. Cross section trend of the $^{49}\text{Ti}(d,x)^{46}\text{Sc}$ reaction (*for target real enrichment).

The $^{49}\text{Ti}(d,x)^{48}\text{Sc}$ reaction cross section is reported in figure 3. The experimental results showed in the graph represent the first measurements of this nuclear reaction. For this reason, the only possible comparison is with TENDL-2025 predictive trend. The agreement between theory and experimental results is quite good over whole energy range covered by this work.

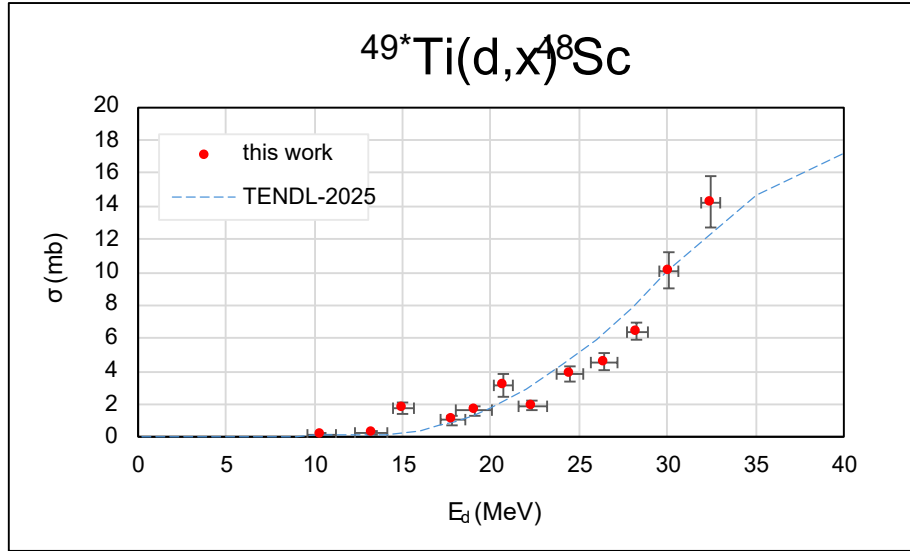


Figure 3. Cross section trend of the $^{49}\text{Ti}(d,x)^{48}\text{Sc}$ reaction (*for target real enrichment).

The last result is about the $^{49}\text{Ti}(d,3n)^{48}\text{V}$ reaction and is reported in figure 4. Again, the obtained cross section values represent the first measurement for this nuclear reaction. However, in this case the TENDL-2025 data do not reproduce the experimental trend. Indeed, experimentally we found an increasing trend over the studied energy range whereas TENDL-2025 foresees a bell-shaped curve with a maximum at about 26 MeV. Additionally, the theoretical curve assumes generally higher values with respect to experimental results, determined by a faster rise of the theory prediction. From a nuclear physics point of view, it could be interesting to continue the experimental investigation at higher energies to get the whole picture of the comparison between the two behaviours. Moreover, even if both the experimental results and the TENDL-2025 prediction correspond to the used enrichment level of the target material ^{49}Ti , the threshold of the two excitation functions seems in disagreement. This can be probably attributed to the activity below the minimum detectable in correspondence of the energies under 17 MeV.

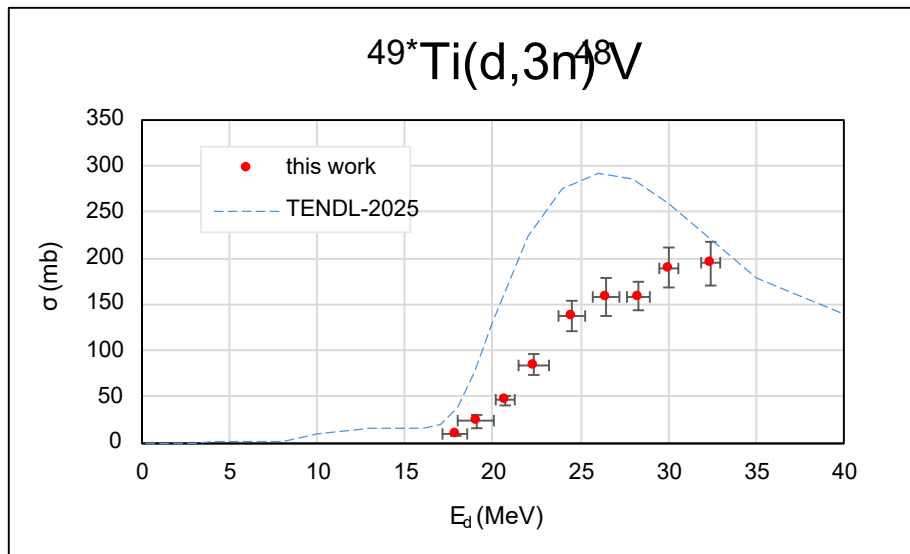


Figure 4. Cross section trend of the $^{49}\text{Ti}(d,3n)^{48}\text{V}$ reaction (*for target real enrichment).

The measured nuclear reactions allowed to investigate the $^{49}\text{Ti}(\text{d},\text{x})$ as a potential route to produce the promising theranostic ^{47}Sc for nuclear medicine applications. For this purpose, the trends of the isotopic contaminants were evaluated in comparison to the ^{47}Sc one. In fact, their presence in the final product can contribute to the radiation dose delivered to patients and medical staff and cannot be removed with the radiochemical purification process. Actually, the validity of $^{49}\text{Ti}(\text{d},\text{x})^{47}\text{Sc}$ as a viable alternative production pathway was already highlighted with a theoretical analysis based on the existing cross section trends in a previous work [Barbaro, 2024], but the necessity of additional experimental measurements was expressed. Thanks to the new results here proposed, which extend the volume of available literature datasets, it is possible to confirm that the production of ^{47}Sc through the $^{49}\text{Ti}(\text{d},\text{x})$ route is truly possible. Specifically, from the trends it can be observed that below 10 MeV ^{47}Sc is produced almost without the co-production of ^{46}Sc and ^{48}Sc , and so a high final radioisotopic purity (>99% for more than 380 h after EOB), one of the parameters to be considered for medical applications, is guaranteed. Concerning ^{48}V , it is supposed that a 100% efficient chemical separation is achievable. Supposing an 80 hrs-irradiation (corresponding to one ^{47}Sc half-life) of a 200 μm thick ^{49}Ti metallic target an activity of 70 MBq/ μA is potentially obtainable at EOB, evaluated starting from experimental cross sections. The yield could be incremented expanding the energy range: for 12 MeV impinging energy the yield grows by 40%, while for 15 MeV the activity produced rises by 60%. However, in these other two different scenarios the productions of ^{46}Sc and ^{48}Sc start (radionuclidic purities of 95.7% and 98.9% at EOB, respectively). Consequently, dosimetric considerations on the contribution to the dose given to a patient due to the presence of these two Sc-contaminants are necessary. A paper discussing the $^{49}\text{Ti}(\text{d},\text{x})$ production route from a dosimetric point of view in many different irradiation conditions is under submission [De Dominicis, 2026]. An alternative to significantly increase the amount of activity is to use very high currents, however this needs a remarkable technological thick targets development to withstand such irradiation conditions.

3 Conclusions

In this study new cross section results for the nuclear reactions $^{49}\text{Ti}(\text{d},\text{x})^{46}\text{Sc}$, ^{47}Sc , ^{48}Sc , and ^{48}V in the energy range 4.4÷32.4 MeV were measured to investigate a possible production route for the theranostic radionuclide ^{47}Sc , particularly promising for the therapy of oncological diseases thanks to its β^- emission. The results about ^{47}Sc and ^{46}Sc were compared with the only previous literature data available [Chen, 1964] showing slightly different trends. In addition, TENDL-2025 library excitation curves [Koenig, 2019] were included for comparison resulting in a good agreement in most cases.

From the analysis of the resulting experimental trends, it was observed that a ^{47}Sc production meeting the purity conditions for medical applications is potentially feasible, at least for deuteron energies below 10 MeV. The maximum reachable activities are not very high, e.g. 70 MBq/ μA supposing an 80 h-irradiation but can be an opportunity to foster at least preclinical studies with this emerging radionuclide. Different irradiation conditions can be taken into consideration to improve the final yield but further investigations for dosimetric implications due to the open of ^{46}Sc and ^{48}Sc production channels are needed. Those insights are discussed in a dedicated paper recently submitted, where also a comparison with the $^{50}\text{Ti}(\text{p},\text{x})$ production route, the only other feasible pathway to obtain ^{47}Sc through enriched Ti targets, is included [De Dominicis, 2026].

In conclusion, it should be mentioned that deuteron cyclotrons are not very spread worldwide. However, the aim of this work is to propose the $^{49}\text{Ti}(\text{d},\text{x})$ as an additional production route, rather than the only one, to be considered as complementary to other methodologies to satisfy the market request that there would be in case ^{47}Sc gets routinely used in clinical procedures.

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Acknowledgements

This work was conducted in the framework of the SPES_MED project which is one of the projects carried out in the context of the LARAMED collaboration. SPES_MED was funded by the National Institute of Nuclear Physics of Italy (recurrent funding) in the mainframe of projects approved by the Interdisciplinary National Scientific Commission 3. The LARAMED project is funded by the Italian Ministry of Education, Universities and Research (Funder DOI: <http://dx.doi.org/10.13039/501100021856>, Award numbers: 0000450-08/01/2014- SCCLA-Y31PREV-A, 0015764-05/05/2016-SCCLA-Y31PREV-A) and it is based at the INFN Legnaro National Laboratories. The activities presented in this work were partially supported by University of Padova (call number: PARD-2023, CUP C93C23004250005).

The cyclotron Arronax is supported by CNRS (Funder DOI: <http://dx.doi.org/10.13039/501100004794>, recurrent funding), Inserm (Funder DOI: <http://dx.doi.org/10.13039/501100001677>, recurrent funding), INCa (Funder DOI: <http://dx.doi.org/10.13039/501100006364>, Award number: INCa-DGOS-Inserm 12558), the Nantes Université (Funder DOI: <http://dx.doi.org/10.13039/501100009371>, recurrent funding), the Regional Council of Pays de la Loire, local authorities, the French government and the European Union. This work has been, in part, supported by a grant from the French National Agency for Research (ANR, Funder DOI: <http://dx.doi.org/10.13039/501100001665>, Award numbers: ANR-11-EQPX-0004, ANR-11-LABX-18-01 and ANR-16-IDEX-007) called “Investissements d’Avenir”, Equipex Arronax-Plus, Labex IRON and ISITE NExT.

The authors would like to thank the GIP ARRONAX team, and especially Arnaud Guertin, for their support. The authors would like to thank also Carlos Rossi Alvarez for the useful scientific discussions.

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